

Florian Huber

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RESEACH INTERESTS

Multi-scale modeling of environmental and social impacts and their spatial-temporal distributions
Non-linear dynamics of stocks and flows in the societal metabolism
Geospatial modelling
Machine Learning / Deep Learning
Unequal exchange
Degrowth and Sufficiency
Decision-making under deep uncertainty and robust decision-making
Computational simulations

EDUCATION

Industrial Ecology, Master of Science

09/22 - 05/25

Leiden University, Delft University of Technology, Netherlands

- Thesis: here Does All the Ore Come From? Predicting Mine-Level Cumulative Production of Copper, Nickel, Zinc Mines in Data-Scarce Regions Using a Geospatial Machine Learning Model
- Relevant courses: Spatial Data Analysis, Static/Dynamic material flow analysis, Methods: Analysing Physical Processes (Environmental extended input-output modelling, Life cycle assessment), Model-based decision making

Industrial Engineering (Design of Functional Materials), Bachelor of Science

09/18 - 03/22

University of Augsburg, Germany

Grade: 1.7

- Thesis: Microstructural Investigation of Process Parameters Affecting Ni90Cr10 and Ni95AlMnSi for the Production of Temperature Sensors in Multi-Material Laser Beam Melting
- Relevant Courses: Resourcegeography, Systematic selection of sustainable materials, Operations research, Production technology, Material Science (I-IV)

ACADEMIC EXPERIENCE

Research Assistant: Multi-Regional Physical Input-Output Models for Copper

04/25 - 10/25

WU *Wien*

- Developing a prototyping a Python-based supply-use table optimization framework to generate multi-regional physical input-output models

Data analytics: Decolonialization of critical raw materials (DCRM)

11/23 - 05/25

Delft University of Technology

Delft

- Development of a Multi-Level Framework that enables both the spatially explicit assessment of the colonial risk of critical raw material extraction concerning Indigenous communities and the tracking of this risk through global supply chains.
- Methods: Geospatial data analysis, Multi regional input-output modeling (MRIOs)
- Tools: Python (Pandas, Geopandas, Rasterio, Pysal)

Teaching Assistant: Earth systems Science and Analysis

09/23 -

Leiden University

Leiden

- Development of interactive Python scripts that introduce students to programming and basic data analytics.
- Tutoring in the course accompanying the programming tutorials

NON ACADEMIC EXPERIENCE

Working Student: Sustainability Consultant

04/23 - 01/24

Nexpirit GmbH

Gütersloh, Germany

- Developing concepts for inventory and impact analysis based on bill of material (BOM) structures
- Conceptualization of inventory assessment trough supply chain queries
- Conceptualization of the integration of derived sustainability indicators (climate change, water consumption) in industrial decision-making (CAD-Software, product life cycle management software (PLM)) in close corporation with Siemens development team.
- System design and conceptualizations in alignment with (EN15804, ISO 14040, ISO 14025)

Working Student: Material Engineering and Sustainability

04/22 - 03/23

Elobau

Leutkirch im Allgäu, Germany

- Material Compliance (REACH, ROHS, CMRT)
- Further development of an integrated Material Management System with a focus on enriching life cycle-relevant environmental data and compliance data for the digital product passport

SKILLS

Languages: German, English (C1)

Software: MS Office (Word, Excel, PowerPoint), Overleaf, Jira, Miro

Methods: Spatial Data Science, Environmental Input-Output Modeling, Static/Dynamic material flow analysis, Machine learning, Exploratory modeling, Scenario modeling

Programming Paradigms: Object-oriented-, Functional- and Parallel programming

Programming Languages: Python (Advanced intermediate), R (Beginner)

Python Packages: Numpy, Pandas, Geopandas, Rasterio, OSMNX, Cartopy, Sklearn, PyTorch, Scipy, Pysheds, PySal, Joblib, Numba, Seaborn, EMA Workbench, Matplotlib

REFERENCES

Victor Maus, Senior Researcher

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03/24 -

Vienna, Austria

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